Algorithm 2 Token-level Perturbations

Require: Number of attack steps k, ICL examples I with length n_1, input ids and input tokens, unlabeled question x_p with length n_2, ground truth y_{gt}, model f parameterized by θ, embedding layer h, selected number of token T_k, judge LLM J_1 for reasoning correctness and J_2 for question correctness.

Ensure: Correct a, Success b, Unattackable c, Wrong d.

Initialize $a, b, c, d \leftarrow 0, 0, 0, 0$ ▷ Initialization for conditions

$(y_p, r_p) \leftarrow \text{argmax}\, P_\theta((y_p, r_p)|I, (x_p))$ ▷ Generate original answer and CoT

if $y_p == y_{gt}$ **then**

  $a \leftarrow 1$

  $q_a, q_c \leftarrow f(I, (x_p, y_p)), f(I, (x_p, y_p, r_p))$ ▷ Get target logits

  $x'_p, r'_p \leftarrow x_p, r_p$

  $x'_r \leftarrow \text{random_insert}(x'_p)$ ▷ Randomly insert tokens into the question part

  $y'_p, r'_p \leftarrow \text{argmax}\, P_\theta((y'_p, r'_p)|x'_r)$

  if $y'_p \neq y_{gt}$ **then**

    $d \leftarrow 1$, **Break**

  else

    if $J_1(r'_p, r_p) == 0$ **then**

      $b \leftarrow 1$, **Break**

    end if

  end if

  insert_indices $\leftarrow$ get_insert_indices(x'_r, x'_p) ▷ Get inserted indices

  for $i \leftarrow 1$ to k **do**

    $q'_c, q'_a \leftarrow f(I, (x_p, y_p)), f(I, (x'_r, y_p, r'_p))$ ▷ Get perturbed logits

    $\mathcal{L}_a, \mathcal{L}_c \leftarrow L_{CE}(q'_a, q_a), L_{CE}(q'_c, q_c)$

    $\mathcal{L}_{opt} \leftarrow \mathcal{L}_c - \lambda \cdot \mathcal{L}_a$ ▷ Loss objective

    $g_{tok} \leftarrow \nabla_e(-\mathcal{L}_{opt})$ ▷ g_{tok} has a shape of [seq_len, vocal_size]

    $rep_{tok} \leftarrow \text{Top1}(g_{tok}, \text{axis} = 1)[\text{insert_indices}]$ ▷ We get the top 1 replacement token for each inserted token on the question part

    $s_{tok} \leftarrow \text{gather}(g_{tok}, \text{input_ids})[\text{insert_indices}]$ ▷ Get gradients of inserted locations on the question part

    $s_{tok} \leftarrow \text{Topk}(s_{tok}, T_k)$ ▷ Select Top-k locations for the optimization goals

    $rank_{tok} \leftarrow rep_{tok}[s_{tok} + n_1]$ ▷ Get the replacement tokens on the selected locations

    $x'_p \leftarrow \text{Swap}(rank_{tok}, \text{input_tokens}, x_p, x'_p, J_2)$ ▷ Alg. 3

  end for

  $y'_p, r'_p \leftarrow \text{argmax}\, P_\theta((y'_p, r'_p)|x'_p)$

  if $y'_p \neq y_{gt}$ **then**

    $d \leftarrow 1$, **Break**

  else

    if $J_1(r'_p, r_p) == 0$ **then**

      $b \leftarrow 1$, **Break**

    end if

  end if

  if $i == k$ **then**

    $c \leftarrow 1$

  end if

end if

return a, b, c, d ▷ Final conditions

of a short natural language description corresponding to a **single-variable linear equation**, making it a benchmark for assessing symbolic reasoning and arithmetic problem-solving skills in language models.

Algorithm 3 Swapping the token

Require: A list of ranked tokens *rank*, input tokens, unperturbed question x_p, perturbed question x'_p, and LLM Judge J_2 for checking the semantic correctness.
Ensure: Perturbed input question x'_p.

```
while True do
    for tok in rank do
        x_tmp ← replace(x'_p, tok)
        if J_2(x_tmp, x_p) == 1 then
            return x_tmp
        else
            x_tmp ← x'_p
        end if
    end for
end while
return x'_p
```

GSM8K (Cobbe et al., 2021) is a large-scale benchmark designed to evaluate the mathematical reasoning abilities of language models. It comprises high-quality, grade-school-level arithmetic word problems, each requiring multi-step reasoning to derive the correct answer. The dataset was curated with a focus on problems that involve **multi-step computations rather than simple numerical lookups** or direct retrieval. GSM8K is widely used to assess Chain-of-Thought (CoT) prompting. It provides a challenging environment where intermediate reasoning steps are crucial for arriving at the correct solution. The dataset has become a standard benchmark in mathematical problem-solving and reasoning research, particularly in the context of LLM fine-tuning and evaluation.

StrategyQA (Geva et al., 2021) is a question-answering benchmark. Unlike traditional datasets, it focuses on questions where the necessary reasoning steps are implicit and must be inferred using a strategy. The dataset comprises 2780 examples, each including a strategy question, its decomposition into reasoning steps, and supporting evidence paragraphs. Analysis indicates that questions in StrategyQA are concise, cover diverse topics, and require a wide range of reasoning strategies. It is widely used in commonsense reasoning.

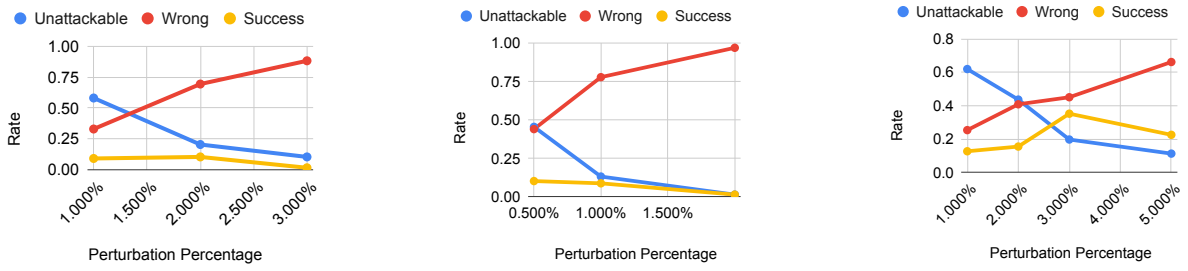
A.3 Algorithms

Algorithm 1 shows the procedure of embedding-level attack, and Alogrithm 2, Algorithm 3 displays the token-level attack procedure. Algorithm 3 shows the steps of replacing the token with the help of an LLM judge checking the semantic correctness of the perturbed question.

A.4 Additional ablation studies

Embedding perturbation percentage ϵ. In Figure 5, for embedding-level attacks, we show UR, WR, and SR on SingleEq, GSM8K, and StrategyQA datasets with varying perturbation percentage ϵ values in $[0.005, 0.01, 0.02, 0.03, 0.05]$ using Llama-3-8B. A larger epsilon leads to a smaller UR and a larger WR. Also, a tradeoff exists for finding the best ϵ for good SRs, which usually falls in the middle. According to the results, we select $\epsilon = 0.02, 0.005, 0.03$ for SingleEq, GSM8K, and StrategyQA respectively, with good SRs.

Unnatural examples for token-level perturbations. We acknowledge that preserving the final answer alone does not guarantee that the semantic meaning of the question remains unchanged. To assess this directly, we performed an additional human evaluation to examine the semantic fidelity of the perturbed inputs. Concretely, we randomly sampled 50 examples from each dataset (SingleEq, GSM8K, and StrategyQA) and asked human annotators to judge whether the meaning of the perturbed question differed substantially from that of the original. Across all datasets, we observe a high degree of semantic preservation, with agreement rates of 98% on SingleEq, 94% on GSM8K, and 88% on StrategyQA (Table 4). These results indicate that, in the vast majority of cases, the perturbations do not materially alter the original question intent, supporting the validity of using LLM-based judges within our evaluation framework.



(a) SingleEq: ϵ.

(b) GSM8K: ϵ.

(c) StrategyQA: ϵ.

Figure 5: Ablation studies on perturbation percentage ϵ of embedding-level attacks using Llama-3-8B on datasets.

Dataset	Agreement
SingleEq	98%
GSM8K	94%
StrategyQA	88%

Table 4: Human evaluation of semantic agreement between original and perturbed questions.

Full results with confidence intervals for Figure 4. The results are shown in Table 5 and Table 6. We observe that a ratio of 0.2 consistently performs well across datasets, with stable performance within the confidence bounds. Increasing the number of steps beyond 5 does not lead to notable improvements, and the differences fall within the margin of error. To ensure efficiency, we use 5 perturbation steps in our main experiments.

A.5 Runtime analysis

In Table 7, we show the runtime for running token-level and embedding-level attacks on three datasets over five model architectures, using NVIDIA H200 GPUs. We see from the table that the token-level attack is more costly compared with the embedding-level attack, because we need to perform semantic evaluations to check whether the meaning of the perturbed question has changed.

A.6 More examples of successful attack

Table 8, 9 and 10 show the complete list of successful attack examples for four error types across three SingleEq, GSM8K, and StrategyQA datasets, with examples from both token-level and embedding-level attacks. From those examples, we observe that MATCHA is capable of creating many successful examples and reveals the sensitivity of different open-source models against input perturbations.

A.7 LLM judge

A.8 Human evaluation of GPT-3.5 judge effectiveness

We perform human verification of GPT-3.5's judgments in Table 11. Specifically, we randomly sample 50 examples from each dataset and manually annotate whether the model's Chain-of-Thought (CoT) reasoning is correct. Human annotations agree with our GPT-3.5 judge around 80% of the time across three datasets (see below). Prior works (Zheng et al., 2023) also note an agreement rate of 70% to 80% between human annotation and LLM judges, indicating that our judge matches the standard set in previous works.